

Sentinel — Solution Presentation

Slide-by-slide content for the submission deck. Each slide gives the headline, the content to put on it, and speaker notes. Export to PDF for submission.

Design guidance: dark slate background (#0F1721), white headings, one accent blue (#2F6FEB), one alert amber (#F0A87C). No clip art. Every number on these slides is measured and reproducible — see

`DOCS/MEASUREMENTS.md` .

Slide 1 — Title

SENTINEL

Statewide CCTV Integration Platform

Model 1 — Central CCTV Registry & GIS Mapping **Model 2** — Unified Viewing Platform with ANPR & Watchlist Alerting

Gujarat CCTV Integration Hackathon 2026 · Category 1 Team: ____ · *Members:* ____

A working platform running against the live Sentinel sandbox grid — 30 cameras onboarded and mapped, a continuous ANPR pipeline validated at 100% on legible footage, automatic watchlist alerting, and route reconstruction across cameras.

Speaker note: Open on the result, not the introduction. One sentence: "Hand us a registration number and we will show you where that vehicle has been, across the grid, with timestamps — because we have been indexing continuously since Thursday."

Slide 2 — The problem, and what wins it

The operational gap

- Cameras are owned by many departments, in many formats, with no common registry
- An investigator asking "where has this vehicle been?" has no way to ask it once
- Watchlist matching, where it exists, is a human watching a screen

What Sentinel does

Onboards	Any camera the catalogue publishes, with location provenance recorded
Watches	Unified live wall, no per-department client
Reads	Continuous ANPR — every plate, every camera, every timestamp, indexed
Correlates	Automatic exact-then-fuzzy watchlist matching
Alerts	Real time, over WebSocket, with severity from the watchlist entry
Reconstructs	Timestamped route across cameras, on a map
Accounts	Every search and export audited with a stated purpose

Speaker note: The last row is the one most teams skip and the one a government evaluator cares about most.

Slide 3 — Architecture



Stack: Python 3.12+ / FastAPI · PostgreSQL 17 + PostGIS + pg_trgm · React + Vite + TypeScript · Leaflet · hls.js · ONNX Runtime (CPU)

Deliberately not built: Kafka, Elasticsearch, Kubernetes, Keycloak. At this scale PostgreSQL does all of it; the scale-out path is designed and documented rather than half-implemented.

Speaker note: Say the last line out loud. Restraint is a competence signal, and the HLD Section 9 shows we know exactly where each of those enters the design.

Slide 4 — Model 1: Registry & GIS

The unpleasant surprise, handled honestly

The sandbox catalogue publishes **only id and name** — no coordinates, no department, no codec.

What we did

- Resolved all 30 camera locations from the real site names in the catalogue
- Strict precedence: hand-verified → Nominatim geocoding (rate-limited, cached, bounded to Gujarat) → district centroid → **null**
- Every camera stores `geo_source` and `geo_confidence`, so provenance travels with the record and is visible in the UI and the report

What we did not do

We did not invent coordinates. An earlier approach placed cameras on an arithmetic grid — a Junagadh-named camera plotted in Surat. That was removed. A camera we cannot locate is reported as unlocated.

GIS features: PostGIS geography points, GeoJSON export, department / status / location-confidence layers, clustered markers, click-through to the live tile.

Speaker note: This slide is about integrity. Cameras now sit in their real districts — Junagadh, Rajkot, Navsari, Kutch — and any evaluator can cross-check a name against the map.

Slide 5 — Model 2: ANPR, and where accuracy comes from

No training. No API key. Nothing leaves the deployment.

Pretrained open-source models (YOLOv9-t detector, cct-s-v2 OCR) on **local CPU**. No frame reaches any third party; no external service sits in the alerting path.

Accuracy comes from two free steps, not from a bigger model

1 • Track-level voting — a vehicle is visible 20–60 frames. Every read votes per character, weighted by the OCR's own per-character confidence, right-aligned on the four-digit serial. Tracks reach 20–35 reads and confidence 1.000.

2 • Indian plate grammar — [2 letters][1-2 digits][1-3 letters][4 digits]. Knowing which positions must be alphabetic makes O↔0, I↔1, S↔5, B↔8, Z↔2, G↔6 deterministically correctable, with the state code validated against the RTO list.

Real example: our OCR returned GJ01AB1234 . Position 2 must be a digit, so the letter O is unambiguously a zero. 7/7 on the correction test cases.

3 • Overlay rejection — the detector offers up burnt-in timestamps, CSITMS-31 , PTZ labels, and the "ADVERTISE HERE" billboard on cam05. Filtered before the index.

Measured on ground truth: 6/6 plates, 0 false positives.

Measured on all 30 sandbox cameras: 29 are wide-area PTZ overviews and yield nothing but signage, correctly rejected. **cam12 is a toll plaza** — and there the same pipeline found a real truck plate **25 times in 100 seconds**, recovering 8 of its 10 characters. It fails only because the plate is **5 px per character**. Seven preprocessing variants were tried; none help, because upscaling cannot restore detail the sensor never captured.

The limit is optics, not software — and the department already has cameras in the right places. See Slide 10.

Speaker note: If asked "did you train a model?" — answer with the sentence in bold at the top. It is a strength, not an admission.

Slide 6 — The engineering decision that matters

These cameras are wide-area night PTZ overviews. A plate is 5–15 px wide.

Full-frame inference at 384 px proposes **nothing** on this grid. Tiled inference on overlapping upscaled regions does find plate-shaped regions — at 15× the CPU cost. On legible footage that difference is 3/6 versus **6/6** plates recovered.

Measured on cam05, 20-core CPU, no GPU	Full frame	Tiled 2×3 @2×
Mean inference	12.3 ms/frame	185.9 ms/frame
Est. concurrent streams per machine	≈251	≈ 26
Plate candidates in a 90 s window	0	8
Ground-truth plates recovered	3/6	6/6

Why this table is the scalability answer

ANPR is not a flat per-camera cost. It is a **policy choice about which cameras carry analytics** — and that choice is now backed by measurement, not assertion.

Tier	Share	Workload
1 — continuous ANPR	5–10%	Highways, border posts, ANPR-sited cameras
2 — event-triggered	20–30%	Urban junctions
3 — registry & view	60–75%	Coverage without analytics cost

Speaker note: Most teams quote one throughput number. Quoting *both* modes, and the zero, is what shows the work. If asked why the live grid yields nothing, go straight to Slide 10 — we measured it stage by stage and the answer is camera siting, not the software.

Slide 7 — Correlation, alerting and route reconstruction

Watchlist matching — exact, then fuzzy

Every detection is checked before its transaction commits: exact match, then `pg_trgm` trigram similarity > 0.7. A match writes an alert and broadcasts it over WebSocket with the watchlist's own reason and severity.

```
SELECT *, similarity(plate_text, :q) AS score
FROM detections WHERE plate_text % :q
ORDER BY score DESC, detected_at;
```

Fuzzy search is not a nicety. ANPR will misread a character, and exact-match-only search fails live on a plate you genuinely detected. Verified: searching `GJ01AB1234` returns the sightings stored as `GJ01AB1234` .

Route reconstruction

Sightings ordered by time → great-circle distance and elapsed time between consecutive pairs → implied speed → **transitions above ~150 km/h flagged and shown**, not silently dropped.

A flagged transition tells an investigator either that a plate was misread or that two vehicles share a similar registration. Both are useful. A system that hides its own uncertainty is harder to trust than one that surfaces it.

Speaker note: Demonstrate the fuzzy search live if there is time. It is the single most persuasive twenty seconds in the demo.

Slide 8 — Security, privacy and accountability

Control	Implementation
Access control	JWT; three roles enforced per route , not just documented
Audit trail	Actor from the verified token , never a request parameter, with stated purpose and case reference
Data minimisation	Plate text, timestamp, camera, plate-region crop — not continuous video
Residency	All inference local; no frame or crop leaves the deployment
Credentials	Environment only; masked in logs; never serialised to the browser
Transport	CSP · X-Frame-Options DENY · nosniff · HSTS · referrer and permissions policies
Rate limiting	10/min on auth, 300/min elsewhere; video segments exempt
DPDP Act 2023	Purpose-bound access, role restriction, minimisation, full access trail

Verified, not asserted: a viewer token gets **403** on the audit trail and on plate search, **200** on the camera registry. The eleventh login attempt in a minute returns **429**.

Closed government databases — handled correctly

VAHAN, SARTHI, eGujCop and NAFIS have no access route for a hackathon team, and we do not pretend otherwise. We define the request/response contract, implement the adapter, mock the endpoint with realistic records, and document exactly what changes when credentials arrive: base URL, auth, rate limiting, cache, audit hook. Nothing above the adapter changes.

That is what integration readiness means, and it survives questioning in a way a claimed live integration would not.

Slide 9 — Scaling to 80,000 cameras

We reproduced the 80,000-camera bottleneck at a scale of eight

Opening N concurrent RTSP connections to the sandbox gateway:

Concurrent	Succeeded	Wall time	Our CPU (20 cores)
2	2 / 2	12 s	4%
4	4 / 4	26 s	4%
6	6 / 6	41 s	5%
8	7 / 8	88 s	4%

Accept times at eight: 4s · 8s · 27s · 33s · 56s · 60s · 63s · 73s — a queue.

Our machine was idle at 4% CPU while the gateway took 73 s to accept the eighth connection. The limit is the single shared ingress, not compute. That is the 80,000-camera problem in miniature, and no amount of faster hardware on our side moves it.

Flat central ingestion does not work, and here is the arithmetic

Quantity	Value
Aggregate ingest bandwidth	≈192 Gbps
Central write rate	≈24 GB/s
Raw volume, 30 days	≈62 PB

Edge-first topology

```
camera → DISTRICT EDGE → REGIONAL DC → STATE CORE
    record          aggregate      registry
    transcode       regional       statewide search
    ANPR inference  search         cross-region routes
    alert generation warm storage  dashboards, audit
           ↓              ↓              ↓
    only metadata,  metadata      metadata only
    alerts, requested + clips
    clips go upstream
```

Only metadata, alerts and requested clips traverse the backbone — reducing the wide-area requirement by **two to three orders of magnitude**.

Where linearity breaks: network fan-in at regional aggregation, metadata hot partitions, storage rebuild times, and scatter-gather cross-region route queries. Each needs load-testing before any statewide commitment.

Why the sandbox ceiling is not our ceiling

	Sandbox	Statewide deployment
Cameras per ingest point	30, one shared gateway	50–200 per district edge node
Who consumes a stream	Every competing team, over the internet	One edge node, on the local network
Across the WAN	Full video to every viewer	Metadata, alerts, requested clips

80,000 cameras over ~400 edge nodes is **200 per node** — ordinary server load, and consistent with our measured 26 concurrent ANPR streams per machine.

Speaker notes: If asked "your wall struggles at nine cameras, how will you do 80,000?" — go straight to the 4% CPU figure. The answer is: *the sandbox is one shared gateway serving every team; no deployment looks like that, and our measurements show the compute side has enormous headroom while the shared ingress is what fails first. That is precisely why the architecture is edge-first — we are watching the argument prove itself at a scale of eight.*

Then: "A measured benchmark of 26 streams with a transparent projection is worth more than an unsupported claim of 80,000."

Slide 10 — What works, what does not

Working, demonstrated, reproducible

✓ 30 cameras onboarded with location provenance ✓ GIS map with department, status and confidence layers ✓ Live video wall — HLS grid plus WebRTC hero tile ✓ Continuous ANPR pipeline with tiled inference, track voting and plate grammar ✓ Exact, partial and fuzzy plate search ✓ Automatic watchlist matching and real-time alerts ✓ Route reconstruction with impossible-transition flagging ✓ Output report — XLSX and PDF ✓ Audit trail with purpose binding ✓ 57 automated tests · 6/6 ground-truth plate accuracy, 0 false positives

Honest limitations

- **The pipeline reads no valid plates from this grid — and we know exactly why.** All 30 cameras scouted. 29 are wide-area PTZ overviews yielding only roadside signage, correctly rejected. **cam12, a toll plaza, is the exception:** 25 detections of one real truck plate in 100 seconds, 8 of 10 characters recovered, failing only on 5 px per character. Seven preprocessing variants change nothing. **The limit is optics, not software** — and the recommendation is specific: toll plazas and checkpoints first. Evidence in [DOCS/evidence/](#).
- Camera coordinates are **geocoded, not surveyed**; accurate to the site, not the pole.
- Department attribution is **inferred** from site type and labelled as such.
- The route is a **sequence of point sightings**; the road-snapped line is an interpolation, returned separately and never presented as evidence.
- Single-node prototype — no HA or DR. The path is designed, not built.

Speaker note: Do not rush this slide. NFSU and DA-IICT reviewers will probe; having already named the weaknesses converts a hostile question into agreement.

Slide 11 — Impact on policing

Today — an investigator with a registration number calls each department, waits for someone to scrub footage, and assembles a timeline by hand over days.

With Sentinel — the number is typed once. Every sighting, every camera, every timestamp, on a map, in seconds, because the index was built continuously rather than searched on demand.

Capability	Operational value
Continuous indexing	The route exists before it is asked for
Automatic watchlist alerts	A stolen vehicle announces itself; nobody watches a wall
Fuzzy search	A single misread character does not lose the vehicle
Cross-department registry	One question, one answer, regardless of who owns the camera
Purpose-bound audit	Surveillance capability with accountability attached

Scope — Models 1 and 2, chosen rather than defaulted to. Models 2, 3 and 4 are competing answers to departmental fragmentation, not a menu: Model 2 connects directly and optimises for the operator, Model 3 federates, Model 4 consolidates. Model 1 is foundational and pairs with exactly one of them. At thirty cameras under one operator, a federation layer would be ceremony around a problem this deployment does not have. Model 3's deliverable also requires federating two *different* VMS platforms and we have access to one; Model 4's requires an 80,000-camera load test on hardware we do not have. We would rather report one measured finding than four partial builds. Full reasoning: HLD §1.3.

Roadmap — federation across departments (Model 3) once a second incompatible VMS is onboarded, tiered analytics rollout by district, VAHAN adapter activation on credential grant, then face recognition and additional analytics on the same edge nodes.

Slide 12 — Verify it yourself

Claim	Command
Throughput, both modes	<code>python anpr_worker.py --benchmark</code>
6/6 accuracy, 0 false positives	<code>python tools/make_sample_feed.py --validate</code>
Grammar and vision logic	<code>python -m pytest tests/ -q</code> → 45 passed
Location provenance	<code>GET /api/v1/ingest/status</code>
Audit trail	<code>GET /api/v1/analytics/audit</code>
Full API surface	<code>http://localhost:8000/api/docs</code>

Repository: __ **Demo videos:** __ **Hosted instance:** __ (test credentials in the submission form)

Every number in this deck is reproduced in `DOCS/MEASUREMENTS.md`, with the command that produces it.